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REVIEW

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High-frequency oscillatory ventilation in children: A systematic review and meta-analysis

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Abstract

Background: High-frequency oscillatory ventilation (HFOV) is an alternative mechanical ventilation mode proposed to reduce ventilator-induced lung injuries and improve clinical outcomes. The aim of this study was to determine the effects of HFOV compared to conventional mechanical ventilation (CMV) when used in children with hypoxemic respiratory failure.

Methods: The literature search was conducted to identify all studies published before December 2020. Eligible studies included a population aged between 28 days and 18 years old, presented original data from randomized controlled trials (RCTs) or observational studies, compared the use of HFOV with CMV. Meta-analyses of the pooled data were performed by using random-effects models with inverse-variance weighting.

Results: A total of 11 studies (2605 cases) were included, most of them evaluating patients with acute respiratory distress syndrome. The mean age of participants was 8.2 months and the mean oxygenation index of those included in the RCTs was 24.4. The effect of HFOV on mortality was not significant, and clinically significant harm or benefit could not be excluded (risk ratio [RR], 0.93; 95% confidence interval [CI], 0.72 to 1.20). No significant difference between groups was found in duration of mechanical ventilation (−2.23; 95% CI, −5.07 to 0.61), treatment failure (RR, 0.28; 95% CI, 0.08 to 1.02), and occurrence of barotrauma (RR, 0.88; 95% CI, 0.39 to 1.99).

Conclusion: The scarce evidence currently available does not allow us to conclude that HFOV has advantages over CMV and further studies are needed to clarify its role in the treatment of acute hypoxemic respiratory failure in children.

KEYWORDS

high-frequency oscillation ventilation, mechanical ventilation, pediatric acute respiratory distress syndrome, respiratory insufficiency, ventilator induced lung injury

1 | INTRODUCTION

Although it is a life-saving therapy, supportive mechanical ventilation (MV) has the potential to cause or aggravate pre-existing lung injuries, especially when inappropriate ventilatory strategies are employed.[1] The damage to the lung inflicted by ventilation, termed "ventilator-induced lung injury" (VILI), is histologically indistinguishable from the diffuse alveolar damage associated with acute respiratory distress syndrome (ARDS).[1] Although there is a lack of evidence supporting the occurrence of VILI in children, the pediatric ARDS (PARDS) is well-recognized as a major clinical entity, with mortality rates of up to 39%.[2–4] With this, lung-protective ventilation strategies have been proposed to avoid, or at least minimize, the occurrence of ventilator trauma, as well as mitigate its important impacts on morbidity and mortality.[5–7]

The basic principles of lung-protective MV consist of limiting plateau pressures (and driving pressure), low tidal volumes, high positive end-expiratory pressure, and acceptance of some degree of hypercapnia and hypoxemia.[8] This strategy has reduced mortality in adults with ARDS when applied during conventional MV (CMV), and scarce evidence suggests benefits also for the pediatric population.[7,9] However, based on similar principles of lung-protective ventilation, another mode of ventilation that can be considered to reduce VILI and improve clinical outcomes is high-frequency oscillatory ventilation (HFOV).[10,11] Although there is a debate about whether it should be used as elective or rescue therapy, this ventilatory mode is considered an alternative in hypoxemic respiratory failure when CMV fails to provide adequate and safe gas exchange.[8]

At present, there is little evidence supporting the use of HFOV in children. The most recent meta-analysis of studies comparing HFOV with CMV concluded that HFOV is not beneficial when used as a lung-protective strategy for adults with ARDS and may even be harmful.[12] Therefore, we conducted a systematic review and meta-analysis to determine the effects of HFOV compared to CMV on mortality, physiological, and clinical outcomes when used for the treatment of children with hypoxemic respiratory failure.

2 | METHODS

This review was performed in accordance with the preferred reporting items for systematic review and meta-analysis statement[13,14] and the Cochrane Handbook.[15] We developed a systematic review protocol with prespecified criteria for study selection, outcome measurements, and analysis. A systematic review protocol was registered in the PROSPERO International Prospective Register of Systematic Reviews database (CRD42020202047).

2.1 | Search strategy

The search was conducted in December 2020. Searched literature databases included PubMed, Embase, and the Cochrane Central Register of Controlled Trials. We also searched for unpublished and ongoing trials in clinicaltrials.gov and controlled-trials.com. Keywords used included combinations of "high-frequency oscillatory ventilation," "high-frequency oscillation ventilation," "pediatrics," "infant," and "child." No language or publication date restrictions were applied. A manual search of the reference lists from the selected articles was also performed. Attempts via email were made to contact the corresponding authors of the included studies to identify missing or confusing data. The complete search strategy is detailed in the Online Supporting Information.

2.2 | Study selection and appraisal

Two reviewers (F.M.D.J. and T.H.d.S.) screened the titles and abstracts independently and in duplicate for potential eligibility. They subsequently read the full texts to determine final eligibility. Discrepancies were resolved through discussion and consensus, and if necessary, the assistance of a third author (M.B.B.) was sought.

Eligible studies fulfilled the following criteria: (1) included population between 28 days and 18 years old; (2) presented original data from randomized controlled trials (RCTs) or observational studies; (3) compared the use of HFOV with CMV for patients with hypoxemic respiratory failure; and (4) contained at least one outcome of interest.

Studies were excluded if they met at least one of the following criteria: (1) were case reports, case series, review articles, or observational studies without control or comparator; (2) included participants receiving extracorporeal membrane oxygenation; (3) included neonates; and (4) compared groups of participants with significant differences in severity of illness (pediatric mortality prediction scores or organ dysfunction scores).

2.3 | Outcomes measures

The primary outcome for the current analysis was in-hospital/30-day/28-day mortality. Secondary outcomes included: (1) duration of MV; (2) length of PICU stay; (3) treatment failure, leading to crossover to the other arm or discontinuation of the study protocol; (4) physiological variables (PaO_2/FiO_2 ratio, OI, mean airway pressure [MAP], $PaCO_2$ and pH); (5) adverse events, such as barotrauma and hemodynamic instability. The definition of each outcome previously mentioned was the same as stated by the authors in each study.

2.4 | Data extraction and study quality

Two reviewers (F.M.D.J. and T.H.d.S.) used a standardized spreadsheet to independently extract data on study design, participants' characteristics, details of ventilation strategies, and study outcomes. No simplifications or assumptions were made. The risk of bias and quality of evidence was independently assessed by the same reviewers. Discrepancies were addressed by rereading the study, in discussions between the two authors, and (if necessary) with the assistance of a third author (M.B.B.).

The Revised Cochrane Risk of Bias Tool (RoB 2) was used to assess the risk of bias in included RCTs.[16] The RoB 2 contains five domains: "randomization process"; "deviations from intended interventions"; "missing outcome data"; "measurement of the outcome"; and "selection of the reported result." One of the following three risks of bias judgments was assigned to each item: "low risk of bias," "high risk of bias," or "some concerns."

The risks of bias in non-randomized studies were assessed with The Risk Of Bias In Non-randomized Studies of Interventions (ROBINS-I) tool.[17] The ROBINS-I contains seven domains, and the risk of bias of each domain was judged as "low risk," "moderate risk," "serious risk," "critical risk," or "no information."

The overall risk of bias in the included studies corresponds to the lowest judgment in any of the domains.[16,17] The Robvis tool was used to create the risk-of-bias assessment plots.[18] The quality of evidence for clinical outcomes was assessed according to recommendations of the GRADE working group.[19]

2.5 | Analysis and synthesis

Descriptive statistics from individual studies were reported using proportions and means with *SD* or median and interquartile range unless otherwise reported. To meta-analyze skewed data, samples means and *SD* were estimated by novel methods described by McGrath et al. using the following website: https://smcgrath.shinyapps.io/estmeansd/.[20]

Meta-analyses of the pooled data were performed by using random-effects models with inverse-variance weighting in Review Manager software, version 5.3.5 (Cochrane Collaboration, 2014), and statistical tests of publication bias using the metabias command in Stata 9.2 (StataCorp, 2006). Dichotomous outcomes were compared using the risk ratio (RR), and continuous outcomes were compared using the mean difference (MD), both with 95% confidence intervals (CI); we considered two-sided $p < .05$ to be statistically significant.

To assess the heterogeneity among the studies for each outcome, both Cochran's Q statistic and the I^2 statistic were used. Heterogeneity was considered to be statistically significant when $p < .05$ or $I^2 > 50\%$.[15] We investigated potential sources of clinical and methodological heterogeneity through a priori subgroup analyses stratified by the type of study (randomized vs. observational) and the overall risk of bias. Sensitivity analyses were conducted to determine the influence of a single study on the overall RR estimates by omitting one study in each turn. In addition, to account for potential changes in clinical practice through the years, a sensitivity analysis including only studies published in the past 15 years was carried out. Trial sequential analysis was carried out by the statistical software, TSA Viewer version 0.9.5.10 beta (Copenhagen: Copenhagen Trial Unit, Center for Clinical Intervention Research, Rigshospitalet, 2016, http://www.ctu.dk/tsa).

Potential publication bias was assessed by visually inspecting the Begg and Mazumdar funnel plots in which the log relative risks were plotted against their standard errors.[21] Publication bias was also assessed by using Begg and Mazumdar's adjusted rank correlation test and Egger et al.'s regression asymmetry test.[21,22]

2.6 | Trial sequential analysis

Meta-analyses may result in type I errors due to random error from the studies included in the meta-analysis which had a small sample size, publication bias, and low quality, and studies whose conclusions tended to be changed by later studies with a larger sample size.[23] For the primary outcome (mortality), we perform a trial sequential analysis (TSA) for included RCTs in the meta-analysis to estimate and correct these limitations and determine whether cumulative evidence is enough reliable. Our assumptions included two-sided testing with a type I error of 0.05, and a type II error of 0.20 (power of 80%). The main results of TSA were shown in a cumulative Z-curve graph, and the monitoring boundary of required information size in the graph was determined according to O′Brien-Fleming α spending function.[24] In addition, the futility boundary was set on the basis of O′Brien-Fleming β-spending function. When the cumulative Z-curve crosses the trial sequential monitoring boundary or enters the futility area, a sufficient level of evidence may have been reached, and no further trials are needed. If the cumulative Z-curve does not cross any of the boundaries, and the required information size has not been reached, there is insufficient evidence to reach a conclusion. TSA was carried out by the statistical software, Trial Sequential Analysis Viewer (TSA Viewer) version 0.9.5.10 beta (Copenhagen: Copenhagen Trial Unit, Center for Clinical Intervention Research, Rigshospitalet, 2016, http://www.ctu.dk/tsa).

3 | RESULTS

3.1 | Literature search

Of the 3342 studies identified using the search strategy, 24 were considered potentially eligible and were assessed in detail in a full-text review. Of these, 11 studies met the inclusion criteria.[25–35] One additional eligible trial was identified from the previously published systematic review and was included.[36] Reviewers agreed on all studies for inclusion. The steps followed to identify the studies meeting the inclusion criteria are summarized in Figure 1.

3.2 | Characteristics of the included studies

Of the included studies, six were RCTs[25,26,28–30,36] and six were observational studies.[27,31–35] A total of 355 participants were enrolled in the RCTs, while 2250 children were evaluated in the

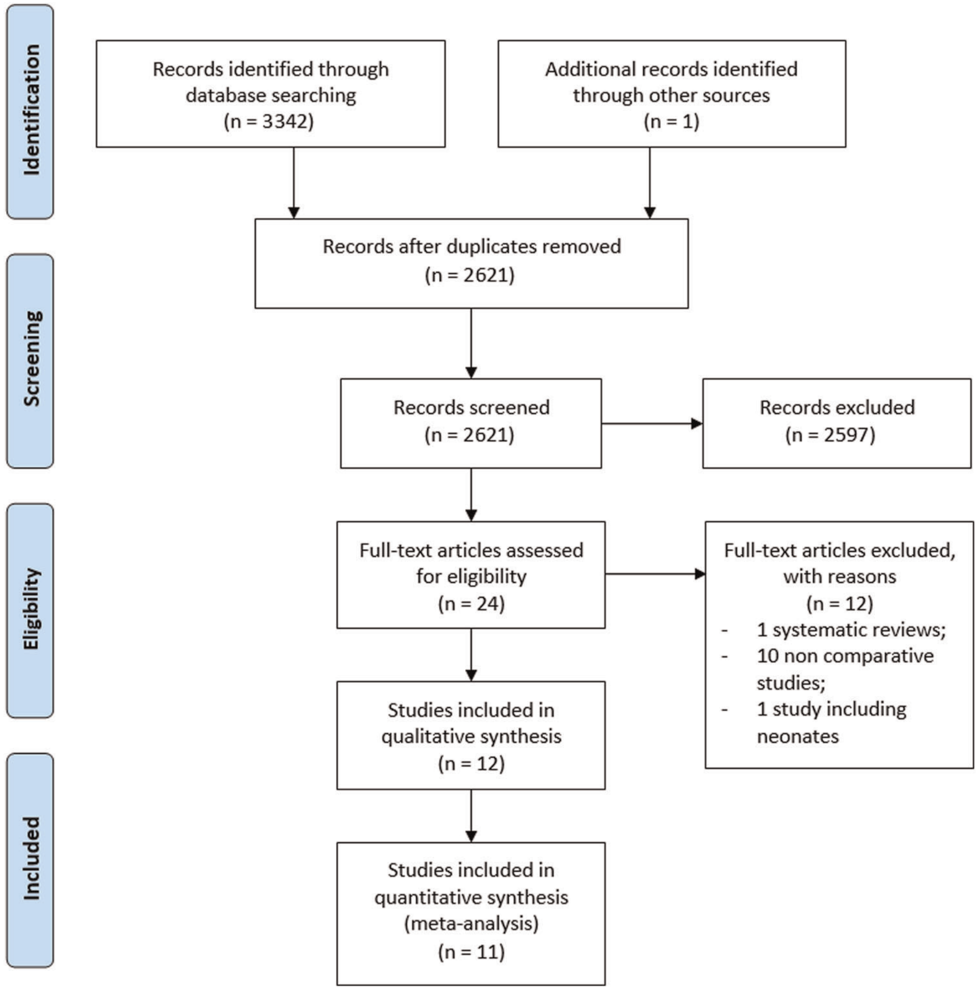


FIGURE 1 PRISMA flow diagram of the selection process. PRISMA, preferred reporting items for systematic review and meta-analysis [Color figure can be viewed at wileyonlinelibrary.com]

observational studies. The overall mean age was 8.2 months. The RCTs enrolled children exclusively diagnosed with ARDS, most of whom were moderate or severe cases (mean OI of 24.4). Most observational studies also evaluated patients with respiratory failure who met the ARDS criteria. All RCTs studied high-frequency oscillation in patients with ARDS who met specific inclusion criteria, as opposed to rescue treatment for refractory hypoxemia. Characteristics of the included studies are shown in Table 1.

Although all RCTs had described the ventilation management strategies used in both groups, they failed to provide full details on lung-protective ventilation strategies used in participants receiving CMV. The ventilation management strategies in the CMV and HFOV groups for each RCT are summarized in Table 2.

3.3 | Risk of bias and quality of evidence

Qualities of evidence (GRADE) for each outcome are presented in Table 3. Details of the risk of bias judgment for primary and secondary outcomes are shown in E-Images 6, 7, 8, 9, and 10.

3.4 | Primary outcome: Mortality

The primary outcome was addressed in 11 studies (five RCTs and six observational studies).[25–29,31–36] Eight studies determined in-hospital mortality,[26–28,31–34,36] two determined 30-day mortality,[25,29] and one study determined 28-day mortality.[35] In the subgroup of RCTs, the pooled RR was 0.93 (95% CI, 0.72–1.20), with no heterogeneity seen among studies ($I^2 = 0\%$, $p = .74$) (Figure 2). However, in the subgroup of observational studies, the pooled RR was 1.37 (95% CI, 0.84–2.22), with significant heterogeneity observed among studies ($I^2 = 85\%$. $p < .001$). The study conducted by Rowan et al. was the main contributor to the heterogeneity observed in its subgroup and in the overall analysis.[34] Rowan et al. evaluated a very specific population of pediatric and young allogeneic hematopoietic cell transplant subjects. After excluding this study, no significant heterogeneity was observed in its subgroup ($I^2 = 31\%$, $p = .23$). When considering only the RCTs published in the past 10 years, the pooled RR was 1.26 (95% CI, 0.87–1.82).[28,29,32–36] A similar result was observed when considering only studies evaluating children with PARDS (RR = 1.03; 95% CI, 0.81–1.31).[25,26,28,29,33–36]

TABLE 1 Characteristics of the included studies

Study	Study design	Participants' demographics: CMV group	Participants' demographics: HFOV group	Participants' characteristics	Inclusion criteria	Outcomes (CMV vs HFOV)
Randomized controlled trials						
Arnold 1994[25] (USA)	Multicenter (5 PICUs); RCT	$N = 29$ 2.5 ± 2.5[a] years 13.1 ± 7.6[a] kg	$N = 29$ 3.1 ± 3.3[a] years 13.6 ± 9.4[a] kg	Children with either ARDS (86%) or pulmonary barotrauma requiring chest tube (14%). Duration of MV before enrollment (h) (CMV vs, HFOV):[a] 80 ± 81 vs. 143 ± 240 ($p > .05$) Baseline PRISM (CMV vs. HFOV):[a] 11 ± 5 vs. 11 ± 5 ($p > .05$) Baseline OI (CMV vs HFOV):[a] 28.5 ± 14.2 vs. 25.5 ± 9.6 ($p > .05$)	**i.** Body weight ≤ 35 kg **ii.** One or both of the following nonexclusive diagnostic categories: acute diffuse lung injury (hypoxemic respiratory failure), associated with poor or decreased lung compliance, resulting in respiratory failure and the need for mechanical respiratory support with positive airway pressure; pulmonary barotrauma. **iii.** One or both of the following measurable conditions: OI of >13, demonstrated by two consecutive arterial blood gas measurements over a 6-h period; radiographic evidence of barotrauma.	30-day mortality: 41% vs. 34% ($p > .05$) Length of MV (days):[a] 29 ± 18 vs. 27 ± 31 ($p > .05$) Barotrauma: 21% vs. 14% ($p > .05$) Hemodynamic instability: 3% vs. 10% ($p > .05$) Neurologic events on study: 7% vs. 3% ($p > .05$) Need for supplemental oxygen at 30 days: 59% vs. 21% ($p = .039$)
Samransamruajkit 2005[26] (Thailand)	Single-center; RCT	$N = 9$ 8.4 ± 4.7[a] years Weight NR	$N = 7$ 2.48 ± 2.06[a] years Weight NR	Children with moderate or severe ARDS Baseline OI (CMV vs. HFOV):[b] 16.4 ± 1.6 vs. 23.9 ± 4.3 ($p > .05$) Baseline PaO_2/FiO2 ratio (CMV vs. HFOV):[b]	Patients were diagnosed as ARDS by AECC definition and met the following entry criteria: **i.** Required PEEP ≥ 5 cmH_2O, **ii.** FiO2 ≥ 0.6 regardless of PEEP level for ≥12 h to maintain	Mortality: 55% vs. 29% ($p = .2$) Length of MV (days):[b] 20 ± 4 vs. 13.6 ± 4 Arterial pH, $PaCO_2$, PaO_2 and PaO_2/FiO_2, and both MAP and OI values markedly improved within the

TABLE 1 (Continued)

Study	Study design	Participants' demographics: CMV group	Participants' demographics: HFOV group	Participants' characteristics	Inclusion criteria	Outcomes (CMV vs HFOV)
				131.7 ± 26.06 vs. 110 ± 30.22 ($p > .05$)	oxygen saturation ≥92% **iii.** OI > 15 for ≥4 h	first 24 h in the HFOV group compared to CMV group (data not shown).
Wenjin 2015[36] (China)	Single-center; RCT	$N = 33$ 7.2 ± 1.8[a] months Weight NR	$N = 30$ 8.1 ± 2.3[a] months Weight NR	Mechanical ventilation time before CMV:[a] 6.85 ± 4.13 days Baseline OI (CMV vs. HFOV):[a] 29.9 ± 2.8 vs. 29.7 ± 2.7 ($p = .473$) Pediatric Critical Illness Score (CMV vs. HFOV):[a] 70.6 ± 12.8 vs. 72.5 ± 13.7 ($p > .05$)	Children with measles pneumonia and severe ARDS, as defined by 2012 Berlin Conference.	Mortality: 45% vs. 33% ($p = .326$) Barotrauma: 24% vs. 17% ($p = .458$) Length of MV (days):[a] 11.03 ± 3.60 vs. 7.97 ± 3.06 ($p = .129$) Length of PICU stay (days):[a] 13.0 ± 7.42 vs. 10.6 ± 6.35 ($p = .215$) OI at 12 h:[a] 28.1 ± 2.6 vs. 24.5 ± 2.3 ($p = .034$)
Samransamruajkit 2016[28] (Thailand)	Single-center; RCT	$N = 9$ 62.5 ± 66.7[a] months 23.1 ± 21.1[a] kg	$N = 9$ 32.9 ± 54.9[a] months 27.5 ± 33.3[a] kg	Children with severe ARDS Participants in CMV group underwent lung volume recruitment maneuver. Duration of MV before enrollment (days) (CMV vs. HFOV):[a] 0.5 ± 1.2 vs. 1.7 ± 0.8 ($p = .8$) Baseline PRISM-3 (CMV vs. HFOV):[a] 11.5 ± 8.1 vs. 12.5 ± 10.7 ($p = .7$) Baseline OI (CMV vs. HFOV):[a] 23.1 ± 9.3 vs. 25.9 ± 11 ($p = .8$) Baseline PaO_2/FiO2 ratio (CMV vs. HFOV):[a] 78.3 ± 27.2 vs. 82.4 ± 24 ($p = .5$)	Patients were diagnosed as ARDS by Berlin definition and met the following entry criteria: **i.** Respiratory failure not fully explained by cardiac failure or fluid overload **ii.** Chest X-ray findings of new infltrates consistent with acute pulmonary parenchymal disease **iii.** Full face-mask, bi-level ventilation or continuous positive airway pressure >5 cm H2O.	Mortality: 11% vs. 11% ($p > .05$) Hemodynamic instability: 0% vs. 0% PaO_2/FiO_2 ratio increase (%) at 1 h:[a] 69 ± 56.8 vs. 138.5 ± 49.7 ($p < .01$) There was no significant difference in PaO_2/FiO_2 ratio at 4 h between both groups (data not shown) There was no significant change in PCO_2 at 1, 4, 6, 12, and 24 h in both groups (data not shown)

(Continues)

TABLE 1 (Continued)

Study	Study design	Participants' demographics: CMV group	Participants' demographics: HFOV group	Participants' characteristics	Inclusion criteria	Outcomes (CMV vs HFOV)
El-Nawawy 2017[29] (Egypt)	Single-center; RCT	N = 100 7.7 (3.0–17.7)[c] months 7.8 (5.0–11.6)[c] kg	N = 100 7.0 (5.0–12.0)[c] months 8.0 (5.33–8.9)[c] kg	Children with diagnosis of moderate or severe ARDS as defined by 2012 Berlin Conference. Baseline PIM-2 (CMV vs. HFOV): 56.55 ± 18.36[a] vs. 55.80 ± 19.02[a] (p > .05) Baseline PaO_2/FiO_2 ratio (CMV vs. HFOV):[c] 110.0 (92.3–143.0) vs. 110.0 (87.3–141.0) (p = .818) Baseline OI (CMV vs. HFOV):[c] 13.0 (10.0–18.3)[c] vs. 16.0 (11.78–22.7)[c] (p = .005)	Included participants fulfilled the AECC criteria for ARDS: **i.** PaO2/FiO2 ≤ 200 mmHg **ii.** Bilateral pulmonary infiltrates on chest radiography without evidence of left atrial hypertension	30-day mortality: 43% vs. 45% (p = .776) Barotrauma: 11% vs. 8% (p = .631) PaO_2/FiO_2 ratio increase (%) at 24 h:[c] 37.5 (26.62–72.03) vs. 67.5 (27.16–109.04) (p = .004) OI decrease (%) at 24 h:[c] 27.4 (20.89–43.5) vs. 41.9 (24.86–52.5) (p = .004) Lengh of MV (days):[c] 5.0 (4.0–7.0) vs. 5.0 (4.0–7.0) (p = .658)
Fioretto 2011[30] (Brasil)	Single-center; RCT with crossover	N = 14 15 (1–83)[d] months Weight NR	N = 14 10.5 (1–40)[d] months Weight NR	Both groups received inhaled nitric oxide. Baseline PaO_2/FiO_2 ratio (CMV vs. HFOV):[a] 111.9 ± 36.9 vs. 123.7 ± 32.9 (p > .05) Baseline OI (CMV vs. HFOV):[a] 20.6 ± 14.3 vs. 16.7 ± 5.6 (p > .05) PRISM (CMV vs. HFOV):[d] 12 (7–16) vs. 10 (3–21) (p > .05)	Hypoxemic respiratory failure was considered present if the patient had two values of PO_2/FiO_2 200 one hour apart, and evidence of moderate to severe respiratory distress shown by dyspnea worsening from baseline, tachypnea, and use of accessory muscles.	Mortality: 0.07% vs. 0.07% (p > .05) PaO_2/FiO_2 ratio at 8 h:[a] 227.9 ± 80.7 vs. 171.21 ± 52.9 (p < .05).
Observational studies						
Dobyns 2002[27] (USA)	Multicenter (7 PICUs); Retrospective study	N = 38 3.3 (0.2–23.0)[d] years Weight NR	N = 12 2.0 (0.1–9.1)[d] years Weight NR	Baseline OI (CMV vs. HFOV):[b] 27 ± 2.1 vs. 32 ± 3.2 (p > .05) Baseline PaO_2/FiO_2 (CMV vs. HFOV):[b] 85.1 ± 4.7 vs. 94.6 ± 40.2 (p > .05) Baseline PRISM (CMV vs. HFOV):[b]	Children with severe acute hypoxemic respiratory failure requiring intubation and mechanical ventilation. Two OI ≥ 15 measured by serial arterial blood gases within 6 h of each other	Mortality: 42% vs. 33% (p = .994) Length of MV (days):[b] 22 ± 4 vs. HFOV 52 ± 28 (p = .098) Treatment failure: 47% vs. 58% (p = .877) Change in PaO_2/FiO_2 ratio (%) at 4 h:[b]

TABLE 1 (Continued)

Study	Study design	Participants' demographics: CMV group	Participants' demographics: HFOV group	Participants' characteristics	Inclusion criteria	Outcomes (CMV vs HFOV)
				11.0 ± 1.0 vs. 8.8 ± 1.4 (p > .05) Mean airway pressure (CMV vs. HFOV):[b] 21 ± 1.0 vs. 26 ± 1.2 (p > .05) Age, sex, primary diagnosis, pediatric risk of mortality scores, and PaO_2/FiO_2 ratio were not different among study groups at the time of enrollment.		16 ± 4 vs. 19 ± 13 (p = .003) PaO2/FiO2 ratio after 24 h: HFOV group has greater improvement than CMV group (data not shown) PaO_2/FiO_2 ratio after 72 h:[b] 148 ± 15 vs. 259 ± 60 (p < .05)
Baird 2005[31] (USA)	Single center; Retrospective study	N = 11 16 ± 46[a] months Weight NR	N = 8 6 ± 7[a] months Weight NR	Risk factors for severe disease were similar in both groups.	All patients with respiratory failure due to respiratory syncytial virus infection admitted to the PICU.	Length of oxygenation therapy (days):[a] 12 ± 7 vs. 30 ± 21 (p < .05) Length of MV (days):[a] 7 ± 4 vs. 18 ± 9 (p < .05) Length of hospital stay (days):[a] 15 ± 9 vs. 41 ± 27 (p < .05) There were no episodes of pneumothorax or pneumomediastinum in any patient in either group, and there were no deaths.
Gupta 2014[32] (USA)	Multicenter (98 PICUs); Retrospective study	N = 882 64.6 ± 68.2[a] months 22.7 ± 22.9[a] kg	N = 882 63.5 ± 66.8[a] months 22.6 ± 23.4[a] kg	Baseline PIM-2 (CMV vs. HFOV):[a] −3.2 ± 1.8 vs. −3.2 ± 1.9 (p = .37) Baseline PRISM-3 (CMV vs HFOV):[a] 11.2 ± 8.6 vs 11.1 ± 8.7 (p = .60)	Patients with the primary or secondary diagnosis of acute respiratory failure or sepsis were included in the study.	Mortality: 8.4% vs. 17.3% (p < .01) Length of MV (days):[b] 14.6 ± 15.4 vs. 20.3 ± 19.7 (p < .01) Length of PICU stay (days):[a] 19.1 ± 18.5 vs. 24.9 ± 22.3 (p < .01)
Guo 2016[33] (China)	Multicenter (2 PICUs); Retrospective study	N = 22 40.50 (48.75)[c] months (p = .002) 14.75 (12.63)[c] kg (p = .001)	N = 26 9.00 (21.75)[c] months (p = .02) 6.55 (5.83)[c] kg (p = .001)	Baseline PRISM-3 (CMV vs. HFOV):[c] 6.59 (3.78) vs. 8.77 (6.04) (p = .150)	Children with severe PARDS, as defined by 2015 Pediatric Acute Lung Injury Consensus Conference (PALICC).	Mortality: 23% vs. 35% (p = .367) Vasoactive agents usage: 64% vs. 96% (p = .007) Barotrauma:

(Continues)

TABLE 1 (Continued)

Study	Study design	Participants' demographics: CMV group	Participants' demographics: HFOV group	Participants' characteristics	Inclusion criteria	Outcomes (CMV vs HFOV)
						9% vs. 15% (p = .674) Length of PICU stay (h):[c] 264 (246) vs. 516 (600) (p = .048) Length of MV (h):[c] 144 (168) vs 444 (456) (p < .001)
Rowan, 2018[34] (USA)	Multicenter (12 PICUs); Retrospective study	N = 46 12.3 (5.2–16.9)[c] years Weight NR	N = 50 9.8 (2.6–15.3)[c] years Weight NR	All but 2 of the subjects had severe pediatric ARDS as defined using the OI or oxygen saturation index criteria established by the Pediatric Acute Lung Injury and Consensus Conference (PALICC)	Pediatric and young adult allogeneic hematopoietic cell transplant subjects requiring invasive mechanical ventilation for critical illness.	Mortality: 76.1% vs. 70% (p > .05) 28-day ventilator-free days:[c] 0 (0-0) vs. 0 (0-11) (p > .05) Length of PICU stay (days):[c] 20.0 (9.0–41.0) vs. 18.0 (10.0–30.0) (p > .05)
Wong, 2020[35] (Asian)	Multicenter (10 PICUs); Retrospective study	N = 118 1.9 (0.6–5.3)[c] years Weight NR	N = 118 2.3 (0.8–5.5)[c] years Weight NR	Baseline PIM-2 (non-HFOV vs. HFOV):[c] 9.4 (5.4–22.4) vs. 8.0 (4.7–27.6) (p = .76) The "HFOV group" was defined by any use of HFOV within the first 7 days of PARDS. The "non-HFOV group" consisted of patients on all other modes of MV (80.1% on CMV and 19.9% on airway pressure release ventilation)	Children with PARDS, as defined by 2015 Pediatric Acute Lung Injury Consensus Conference (PALICC).	28-day mortality: 16.9% vs. 32.2% (p = .01) 28-day ventilator-free days:[c] 4.0 (0.0, 17.8) vs. 4.0 (0.0, 16.0) (p = .29) 28-day intensive care unit-free days:[c] 4.0 (0.0, 15.8) vs. 0.0 (0.0,11.0) (p = .03)

Abbreviations: AECC, American-European Consensus Conference; ARDS, acute respiratory distress syndrome; CMV, conventional mechanical ventilation; FiO_2, fraction of inspired oxygen; HFOV, high-frequency oscillatory ventilation; MAP, mean airway pressure; MV, mechanical ventilation; NR, not reported; OI, oxigenation index; PARDS, pediatric acute respiratory distress syndrome; PEEP, positive end-expiratory pressure; PICUs, pediatric intensive care unit; $PaCO_2$, partial pressure of carbon dioxide; PaO_2, partial pressure of oxygen; PIM, pediatric index of mortality; PRISM score, pediatric risk of mortality score; RCT, randomized controlled trial.

[a]Mean ± *SD*.

[b]Mean ± *SE*.

[c]Median (interquartile range).

[d]Median (range).

A random-effects model was applied in the trial sequential analysis for the included RCTs. The required information size to demonstrate or reject a 12.9% relative risk reduction (low-bias risk trial estimate) of deaths was 2526 (Figure 3). The cumulative *Z*-curve did not cross either the boundary for benefit or the required information size, indicating that the evidence was not sufficiently conclusive and further studies are needed.

Evidence of publication bias was found in the funnel plot and Egger's test ($p = .020$), but not in the Begg's test ($p = .727$).

3.5 | Secondary outcomes

3.5.1 | Duration of MV

Ten studies reported the duration of MV.[25–27,29,31–36] Four studies presented data related only to survivors,[25,26,29,31] while two presented data of the entire sample (survivors plus non-survivors)[33,34] and three did not specify this aspect.[27,32,36] Wong et al.[35] presented the duration of MV as 28-day ventilator-free days, which precluded its inclusion in the meta-analysis. In the subgroup of RCTs, the pooled analysis showed no significant difference in duration of MV between groups (MD, −2.23; 95% CI, −5.07 to 0.61; $I^2 = 79\%$, $p = .002$, E-Image 1).

3.5.2 | PICU length of stay

Five studies reported length of stay in the PICU and the results are showed in Table 1.[29,32–34,36] However, most studies did not specify whether the data provided refer only to survivors or whether non-survivors were also considered. Because of the paucity of data from RCTs and the lack of methodological details, the pooled analysis was not performed (E-Image 2).

3.5.3 | Treatment failure

Five studies reported patients who were crossed over from one group to another due to treatment failure.[25–29] Three studies defined the criteria for treatment failure.[25,27,29] In general, treatment failure was declared when patients presented refractory hypoxemia, severe hypercarbia with pH less than 7.20, persistent air leak, or shock in spite of adequate preload and inotropic support. In the subgroup of RCTs, HFOV did not significantly reduce the likelihood of treatment failure compared to CMV (RR, 0.28; 95% CI, 0.08–1.02; $I^2 = 39\%$, $p = .18$; E-Image 3).

3.5.4 | Barotrauma

The occurrence of pneumothorax was addressed in seven studies.[25,26,28,29,31,33,36] No significant difference was found in the incidence of pneumothorax between HFOV and CMV in the subgroup of RCTs (RR, 0.88; 95% CI, 0.39–1.99; $I^2 = 33\%$, $p = .21$; E-Image 4).

3.5.5 | Hemodynamic effects

Six studies provided some information on the hemodynamic status of the patients included.[25,28,30,33,34,36] Arnold et al. reported a similar incidence of hemodynamic deterioration in HFOV and CMV groups (3/29 vs. 1/29, respectively, $p > .05$).[25] Fioretto et al., Baird et al., and Wenjin et al. did not observe hemodynamic instability in any of the studied groups.[30,31] Rowan et al. reported similar use of vasopressors/inotropes agents between groups, while Guo et al. observed that patients undergoing HFOV were exposed to vasoactive agents more frequently than those in the CMV group (96.15% vs. 63.64%, $p = .007$).

3.5.6 | Physiological outcomes

Seven studies compared oxygenation (OI or PaO_2/FiO2 ratio) between groups.[25–30,36] Most studies showed a significantly greater improvement in oxygenation in the HFOV group compared to the CMV group (Table 1). There were no significant changes in PCO_2 in the few studies in which this variable was addressed.[25,28–30] Arnold et al.[25] observed a higher MAP in the HFOV group during the 72 h of study. Unfortunately, due to the great variability in physiological measures among studies, these outcomes could not be examined via meta-analysis.

4 | DISCUSSION

In this systematic review and meta-analysis of studies involving children with hypoxemic respiratory failure, the use of HFOV was not associated with significant differences in mortality when compared to CMV, and clinically significant harm or benefit could not be excluded. In addition, its use did not alter the duration of MV nor the length of PICU stay. Although the vast majority of studies have reported improvement in oxygenation, this does not seem to result in benefits for important clinical outcomes. In fact, our results do not support the routine use of HFOV as an early strategy of lung-protective ventilation in children with ARDS. However, this meta-analysis also does not allow us to conclude that there is no role for HFOV in the low tidal volume ventilation era.

HFOV is an alternative MV mode that maximizes alveolar recruitment, avoids overdistention, and utilizes volumes that are less than dead space, thereby preventing VILI.[37] It has been utilized safely for over 30 years in the neonatal setting and was rigorously evaluated in several RCTs comparing HFOV with other ventilation modes. Even so, its efficacy in this specific population is questionable, since many studies revealed no reduction in the risk of

TABLE 2 Details of high-frequency oscillatory ventilation and conventional mechanical ventilation in the randomized controlled trials included

	Conventional mechanical ventilation					High-frequency oscillatory ventilation			
Studies	Mode	Tidal volume	Permissive hypercapnia	Permissive hypoxemia	PEEP titration strategy (cmH_2O)	Frequency (Hz)	Mean paw (cmH_20)	ΔP (cmH_2O) titration paramenter or goal	Criteria for transition to CMV
Arnold 1994[25]	Pressure limited	NR	YES (hypercarbia was tolerated as long as the arterial Ph was >7.25)	YES (oxygenation goal was $SaO_2 \geq 90\%$ with $FiO_2 < 0.6$)	The PEEP was increased by 2 to 4 cmH_2O increments until oxygenation was improved or until there were signs of compromised cardiac function.	5–10	4–8 above MAP on CMV then titrated to maintain $SaO_2 \geq 90\%$ with an FiO_2 of ≤0.6	Achieve adequate chest wall movement by subjective impression and/or objective monitoring of transcutaneous PaCO2	MAP 18 cmH2O, tolerating suctioning
Saramsamruajkit 2005[26]	Time-cycled or Pressure-controlled	Low tidal vo-lume-s	YES (hypercarbia was tolerated as long as the arterial Ph was >7.25 and PaCO2 between 45–65 mmHg)	YES (oxygenation goal was SaO2 > 92% with $FiO_2 < 0.6$)	NR	4-10	2–3 above MAP on CMV then titrated to maintain $SaO_2 > 92\%$ or $PaO_2 > 70$ torr	10 above peak inspiratory pressure during CMV	MAP 18 cmH_2O, tolerating suctioning
Wenjin 2015[36]	Pressure-controlled or Pressure regulated volume control	4-6 ml/kg	NR	YES (oxygenation goal was $SaO_2 > 88\%$)	Optimal PEEP between 5–10 cmH2O	9–12	18–25	Achieve vibration from chest wall to groin	FiO2 < 0.4, MAP < 14 cmH2O; radiographic improvement; pH 7.35-7.45, $PaCO_2$ 35-45 mmHg, PaO_2 80-250 mmHg
Saramsamruajkit 2016[28]	NR	6–8 ml/kg	NR	NR	NR	Setting according to age	5–8 above MAP on CMV	The initial pressure amplitude was set at 3x MAP of CMV	NR

TABLE 2 (Continued)

	Conventional mechanical ventilation					High-frequency oscillatory ventilation			
Studies	Mode	Tidal volume	Permissive hypercapnia	Permissive hypoxemia	PEEP titration strategy (cmH_2O)	Frequency (Hz)	Mean paw (cmH_20)	ΔP (cmH_2O) titration paramenter or goal	Criteria for transition to CMV
El-Nawawy 2017[29]	NR	5–8 ml/kg	YES (hypercarbia was tolerated as long as the arterial Ph was ≥7.2 and $HCO_3 \geq 19$)	YES (oxygenation goal was $SaO_2 > 90\%$ with $FiO_2 < 0.6$)	Titrated till achieve best oxygenation and/or best respiratory system compliance values.	5–12	3–5 above MAP on CMV then titrated to maintain $SaO_2 > 90\%$ with an FiO_2 of <0.6	Amplitudes were set to assure tidal volumes 1.5-3 ml/kg either directly, or by using the volume guarantee option in the HFP device mode, assuring adequate chest wiggling	SaO_2 maintained above 95% with MAP of 12–15 cmH_2O and FiO2 = 0.4 for 6 h
Fioretto 2011[30]	Pressure-controlled assist/control ventilation	5–7 ml/kg	YES	YES (oxygenation goal was $SaO_2 > 90\%–92\%$ with FiO2 < 0.6)	PEEP was gradually increased to reach MAP close to those used in HFOV	10 Hz for infants (1 year old), and 5–8 Hz for older children	2–4 cmH2O above MAP on CMV then titrated to maintain FiO2 of < 0.6	Pressure amplitude was set to attain perceptible chest wall motion, and to reach targeted $PaCO_2$, as in CMV	

Abbreviations: CMV, conventional mechanical ventilation; FiO_2, fraction of inspired oxygen; MAP, mean airway pressure; NR, not reported; $PaCO_2$, partial pressure of carbon dioxide; PaO_2, partial pressure of oxygen; PEEP, positive end-expiratory pressure; SaO_2, arterial oxygen saturation.

TABLE 3 GRADE summary of findings comparing high-frequency oscillatory ventilation to conventional mechanical ventilation for children with hipoxemic respiratory failure

Outcomes	Anticipated absolute effects[a] (95% CI)		Relative effect (95% CI)	No of participants (studies)	Certainty of the evidence (GRADE)[b]
	Risk with CMV	Risk with HFOV			
Mortality (randomized trials)	422 per 1000	388 per 1.000 (296–494)	RR 0.92 (0. 07–1.17)	355 (5 studies)	⊕⊕◯◯ LOW
Mortality (observational studies)	156 per 1000	237 per 1.000 (151–351)	RR 1.52 (0.97–2.25)	2250 (6 studies)	⊕◯◯◯ VERY LOW
Treatment failure (randomized trials)	265 per 1000	74 per 1.000 (29–170)	RR 0.28 (0.11–0.57)	292 (4 studies)	⊕◯◯◯ VERY LOW
Treatment failure (observational studies)	308 per 1000	410 per 1.000 (157–721)	RR 1.33 (0.51–2.34)	342 (1 study)	⊕◯◯◯ VERY LOW
Barotrauma (randomized trials)	111 per 1000	96 per 1.000 (38–222)	RR 0.86 (0.34–2.00)	355 (5 studies)	⊕⊕◯◯ LOW
Barotrauma (observational studies)	61 per 1.000	105 per 1.000 (19–416)	RR 1.73 (0.31–6.86)	67 (2 studies)	⊕◯◯◯ VERY LOW
Duration of Mechanical ventilation (randomized trials)	-	MD 2.23 days fewer (5.07 fewer–0.61 more)	-	220 (4 studies)	⊕⊕◯◯ LOW
Duration of Mechanical ventilation (observational studies)	-	MD 5.63 days more (0.52 fewer–11.79 more)	-	2248 (6 studies)	⊕◯◯◯ VERY LOW
PICU length of stay (randomized trials)	-	MD 2.4 days fewer (5.8 fewer–1 more)	-	63 (1 study)	⊕⊕◯◯ LOW
PICU length of stay (observational studies)	-	MD 3.42 days more (5.83 fewer–8.57 more)	-	1908 (3 studies)	⊕◯◯◯ VERY LOW

Abbreviations: CI, confidence interval; CMV, conventional mechanical ventilation; HFOV, high frequency oscillatory ventilation; MD, mean difference; OR, odds ratio; PICU, pediatric intensive care unit.

[a]The risk in the intervention group (and its 95% CI) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI).

[b]GRADE Working Group grades of evidence: high certainty - we are very confident that the true effect lies close to that of the estimate of the effect; moderate certainty - the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different; low certainty—the true effect may be substantially different from the estimate of the effect; very low certainty—the true effect is likely to be substantially different from the estimate of effect.

bronchopulmonary dysplasia, neurological insults, or death.[38] In adults with ARDS, enthusiasm for the use of HFOV has reduced considerably after two large RCTs have shown disappointing results.[10,11] The OSCAR trial showed no difference in 30-day mortality,[11] whereas the OSCILLATE trial was terminated early by the steering committee because of significantly higher in-hospital and 60-day mortality in the HFOV group.[10] Although these studies are subject to several pertinent criticisms, their results bring us into question: “If HFOV has so many theoretical advantages over CMV, why does it fail to reduce mortality?” However, the answer to this question may lie in another question: “Is mortality an appropriate outcome for assessing the effectiveness of HFOV?”

Before attempting to answer these questions, it must be kept in mind that most ARDS patients die from multiple organ dysfunction syndrome (MODS) rather than refractory hypoxemia, which is responsible for a minority of deaths.[39,40] This may explain, at least partially, why the significant improvement in oxygenation observed with HFOV does not translate into a clinical benefit. In fact, improvement in oxygenation does not necessarily indicate reduced severity of lung injury or resolution of the ARDS etiology. With this, the greater advantage of HFOV over CMV may rely on preventing the development of VILI by limiting overdistension (barotrauma and volutrauma) and minimizing the cyclic opening and closing of alveoli (atelectrauma). It is believed that VILI can increase pulmonary and

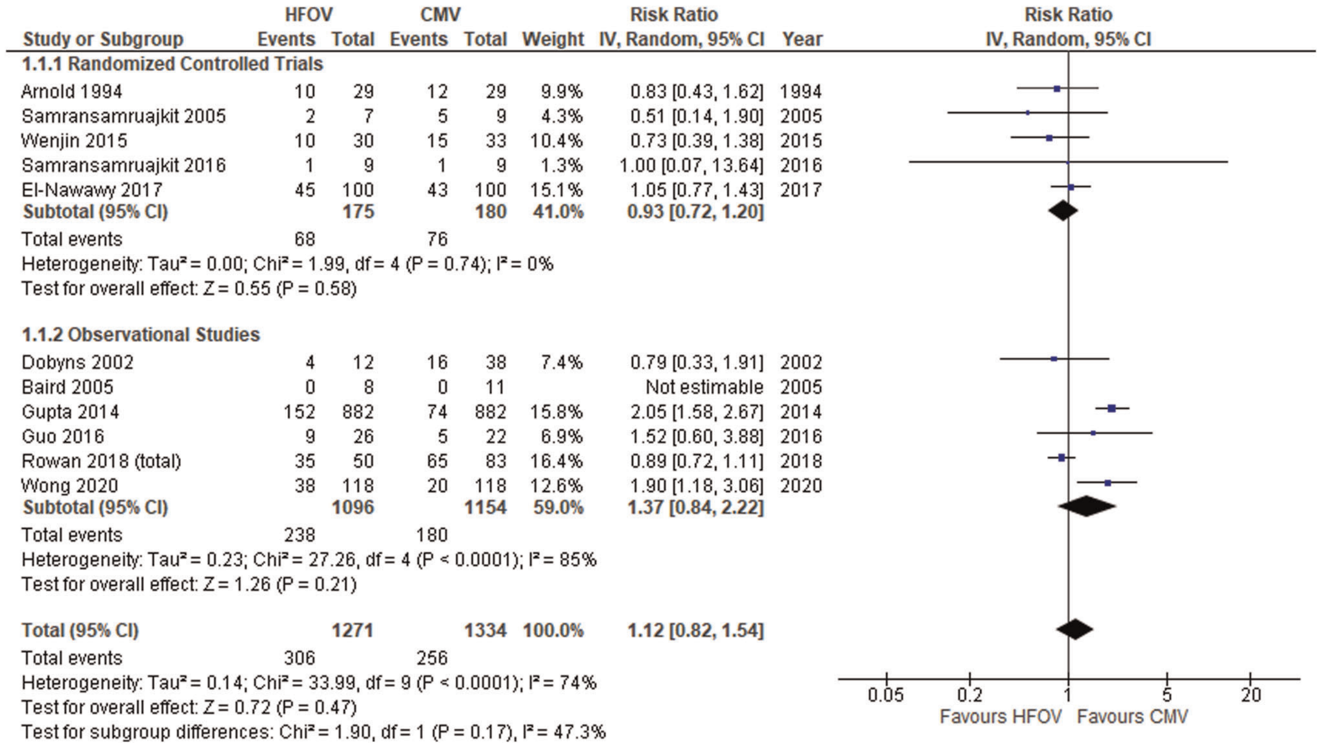
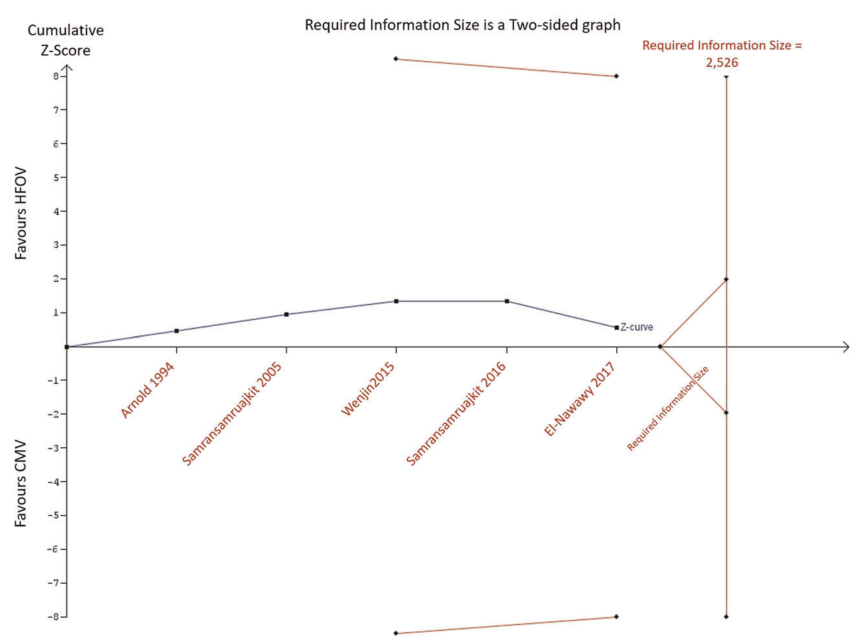


FIGURE 2 Effect of high-frequency oscillatory ventilation (HFOV) on mortality in children with hypoxemic respiratory failure. Subgroup of randomized controlled trials included only patients diagnosed as moderate or severe acute respiratory distress syndrome. "Events" columns show the number of deaths, and "Total" columns show the number of subjects in each group. Horizontal bars indicate 95% confidence intervals (CIs). Vertical dashed line indicates position of mean pooled treatment effect. CMV, conventional mechanical ventilation; I^2, heterogeneity index; IV, inverse variance

FIGURE 3 Trial sequential analysis showing that the cumulative Z curve (blue line) has not crossed the monitoring boundary curve (top and bottom red lines) for benefit, as well as has not yet reached the required information size [Color figure can be viewed at wileyonlinelibrary.com]



systemic cytokine levels, leading to the development of systemic inflammatory response syndrome (SIRS) and, potentially, MODS.[41,42] In animal models, HFOV has shown to reduce inflammatory, histopathological, and oxidative lung injuries when compared to conventional lung-protective MV.[43] Perhaps the incidence of VILI, SIRS, and MODS are more appropriate outcomes to be assessed in further studies evaluating ventilatory modes instead of oxygenation or mortality. In fact, a substantial reduction in mortality of pediatric ARDS was observed over time, which may be related to improvements in general pediatric critical care support and the use of lung-protective ventilatory strategies.[2] Reducing mortality nowadays may have become a herculean task for either HFOV or any other ventilatory mode.

In this meta-analysis, we decided to include observational studies in addition to RCTs. This decision was made due to the known scarcity of pediatric RCTs on this topic. We believed that non-randomized studies can provide valuable data since they are analyzed and interpreted cautiously in light of their limitations. Importantly, observational studies did not allow to strictly control the ventilatory strategies applied in the evaluated groups, which were better addressed in the RCTs. Therefore, we were careful to analyze the included studies in different subgroups according to their study design. As expected, evidence from the subgroup of observational studies had a weaker quality and greater potential biases compared to the subgroup of RCTs. Interestingly, the worst results with the use of HFOV were observed in the subgroup of observational studies. Two retrospective studies of large databases showed significantly higher mortality in the HFOV group.[32,35] Both studies used statistical approaches (propensity score or genetic matching) to account for covariates and minimize biases. Even so, the quality of the evidence in these studies is not comparable to that obtained from RCTs and several criticisms were made to its methods.[44–47] Nevertheless, they highlight the potential harm caused by inappropriate use of HFOV and the need for more RCTs on this topic. HFOV should be employed in its most optimal fashion, taking into consideration the physiological properties of specific lung diseases to determine the best oscillator settings.[48] Currently, the pediatric intensive care community is looking forward to the Prone and Oscillation Pediatric Clinical Trial (PROSpect), which may clarify the role and optimal management of HFOV for PARDS.

Although not statistically significant, the greatest benefit of using HFOV found in this meta-analysis was for treatment failure in the subgroup of RCTs (RR 0.28; 95% CI, 0.08–0.02). However, this result needs to be cautiously interpreted. Two studies in this subgroup did not previously establish the criteria for treatment failure.[26,28] Also, it is important to note that El-Nawawy et al.[29] applied different criteria for treatment failure in each studied group. This may have contributed to the better results obtained in the HFOV group since refractory hypoxemia and barotrauma were considered criteria only for the CMV group. Therefore, the evidence for this outcome was considered to be of low quality and with a high risk of bias.

Some limitations of the present meta-analysis need to be pointed out. First, the number of participants included in this meta-analysis did not reach the required sample size calculated by the TSA. Also, the cumulative *Z*-curve did not cross the trial sequential monitoring boundary or enters the futility area, which means that a sufficient level of evidence was not reached to allow strong recommendations. Second, some included studies failed to provide a clear and detailed description of the ventilation strategy employed in the groups evaluated. Thus, the results presented in the included studies and, consequently, in the present meta-analysis, may have been substantially influenced by the operator rather than the mode of ventilation itself. Thirdly, as previously discussed, we decided to analyze data from observational studies. However, the pooled RR obtained from such studies were presented separately and the risk of bias of each study for each outcome was carefully assessed, as well as the quality of evidence in accordance with GRADE recommendations. Fourthly, there were some variations in the protocol of HFOV and CMV used in the included studies. This limitation is certainly more important for the subgroup of observational studies. Finally, many studies fail to provide the time of MV before initiation of HFOV and this variable could not be adequately analyzed.

5 | CONCLUSION

The scarce evidence currently available does not allow us to conclude that HFOV has advantages over CMV and further studies are needed to clarify its role in the treatment of acute hypoxemic respiratory failure in children. Therefore, it should not be routinely used in this population as a primary ventilation strategy in place of conventional lung-protective ventilation until further RCTs using standardized approach evidence its advantages and safety.

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AUTHOR CONTRIBUTIONS

Tiago H. de Souza: conceptualization; methodology (lead); data curation and statistical analysis (lead); project administration (lead); writing review & editing (lead). **Fernanda M. D. Junqueira:** data curation (equal); methodology (supporting); project administration (equal); writing original draft (equal). **José A. Nadal:** data curation (equal); writing original draft (equal). **Marcelo B. Brandão:** data curation (equal); writing original draft (equal). **Roberto J. N. Nogueira:** writing original draft (equal); writing review & editing (equal).

DATA AVAILABILITY STATEMENT

The protocol for the systematic review and dataset (RevMan 5.0 format) are available by request to the corresponding author.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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